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DEC 18 2019

U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387(388)/2019-002-01
UNIT 1 LICENSE NO. NPF-14
UNIT 2 LICENSE NO. NPF-22
PLA-7835**

**Docket No. 50-387
and 50-388**

Attached is a supplement to Licensee Event Report (LER) 50-387(388)/2019-002-00 that reported an event involving the "C" Emergency Service Water pump that was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications and 10 CFR 50.73(a)(2)(v) as an event or condition that could have prevented fulfillment of a safety function.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.

A handwritten signature in black ink, appearing to be "K. Cimorelli", written over a horizontal line.

K. Cimorelli

Attachment: LER 50-387(388)/2019-002-01

Copy: NRC Region I
Ms. S. Goetz, NRC Project Manager
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Mr. M. Shields, PA DEP/BRP



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Susquehanna Steam Electric Station Unit 1	2. Docket Number 05000387	3. Page 1 OF 3
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4. Title
Inoperability of "C" Emergency Service Water Pump

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
08	28	2019	2019	002	01	12	18	2019	Susquehanna Steam Electric Station Unit 2	05000388
									Facility Name	Docket Number
									N/A	05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level 100	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☒ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact

C. E. Manges, Jr., Senior Engineer – Nuclear Regulatory Affairs

Telephone Number (Include Area Code)

570-542-3089

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES
E	BI	P	Flowsolve (previously Byron Jackson)	Yes					

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On August 28, 2019 at approximately 17:40, during Inservice Testing (IST) of the "A" loop of Emergency Service Water (ESW), the differential pressure for the "C" ESW pump (0P504C) was found to be 88.875 psid, which was below the established IST acceptance criteria. Based on the test results, the "C" ESW pump was in the IST Required Action Range, which requires the pump to be declared inoperable. At the time of discovery of this condition, Susquehanna was already in Technical Specification (TS) 3.7.2 Condition C for "A" loop maintenance. On August 29, 2019, the pump lift setting was adjusted and the "C" ESW pump was retested. The retest resulted in an acceptable differential pressure of 91.9 psid, which was in the Alert Range and no longer in the Required Action Range. TS 3.7.2 Condition C was exited on August 29, 2019 at approximately 15:45. On October 04, 2019, Susquehanna completed replacement of the "C" ESW pump.

The direct cause was determined to be separation of the lower pump casing flange (bottom column to top case) resulting in a loss of differential pressure. The apparent cause was determined to be corrosion of the carbon steel bolting in raw water. This was accelerated by galvanic interactions with silt, Copper never-seize, and alloy steel. Completed corrective actions include replacement of the "C" ESW pump. Planned corrective actions include a review of the appropriate preventive maintenance (PM) frequency for pump replacement.

Based on available information, the condition is considered to have existed for longer than allowed by TS 3.7.2. The condition is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications and 10 CFR 50.73(a)(2)(v) as an event or condition that could have prevented fulfillment of a safety function.

There were no actual safety consequences associated with the condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station Unit 1	05000-387	2019	002	01

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On August 28, 2019 at approximately 17:40, during Inservice Testing (IST) of the "A" loop of Emergency Service Water (ESW) [EIS System Code: BI], the differential pressure for the "C" ESW pump (0P504C) [EIS Component Code: P] was found to be 88.875 psid, which was below the established IST acceptance criteria. Based on the test results, the "C" ESW pump was in the IST Required Action Range, which requires the pump to be declared inoperable. At the time of discovery of this condition, Susquehanna was already in Technical Specification (TS) 3.7.2 Condition C for both units in support of maintenance activities on the "A" loop of ESW.

On August 29, 2019, the pump lift setting was adjusted and the "C" ESW pump was retested. The retest resulted in an acceptable differential pressure of 91.9 psid, which was in the Alert Range and no longer in the Required Action Range. TS 3.7.2 Condition C was exited on August 29, 2019 at approximately 15:45.

On October 04, 2019, Susquehanna completed replacement of the "C" ESW pump. The removed pump had notable degradation, including flange separation and bolting corrosion. The pump had been installed for slightly more than 10 years prior to replacement. Other pumps exposed to the same raw water environment include the "A", "B", and "D" ESW pumps and the "1A", "1B", "2A", and "2B" Residual Heat Removal Service Water (RHRSW) pumps [EIS System/Component Code: BI/P]. Of these pumps, the "2B" RHRSW has been installed for the longest period of time (slightly more than 11 years).

Based on available information, the condition is considered to have existed for longer than allowed by TS 3.7.2. The condition is therefore being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications. The "B" loop of ESW was determined to have been out of service and inoperable for ten minutes for system valve testing on September 29, 2019; as a result, the condition is also being reported in accordance with 10 CFR 50.73(a)(2)(v) as an event or condition that could have prevented fulfillment of a safety function.

CAUSE OF EVENT

The direct cause was determined to be separation of the lower pump casing flange (bottom column to top case) resulting in the flange o-ring protruding and subsequent loss of differential pressure. The apparent cause was determined to be corrosion of the carbon steel bolting in raw water. This was accelerated by galvanic interactions with silt, Copper never-seize, and alloy steel.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Susquehanna Steam Electric Station Unit 1		05000-387	YEAR	SEQUENTIAL NUMBER	REV NO.
			2019	002	01

ANALYSIS/SAFETY SIGNIFICANCE

An engineering evaluation was completed to analyze having only the 'A' ESW pump available. The engineering evaluation concluded that, with only the 'A' ESW pump in operation, all ESW users could have removed the required heat load and all safety related ESW users would have performed their safety related functions. Based on the Engineering evaluation, the condition involving the 'C' ESW pump not being able to perform its safety function did not result in a safety system functional failure. Accordingly, this event will not be counted as a safety system functional failure in the Reactor Oversight Process Performance Indicators.

CORRECTIVE ACTIONS

Key corrective actions include the following:

1. Susquehanna completed replacement of the "C" ESW pump on October 04, 2019.
2. Susquehanna will determine the appropriate preventive maintenance frequency for pump replacement.

In addition to the actions above, an extent of condition action will be taken to inspect the 2P506B RHRSW Pump to determine if similar corrosion is observed.

COMPONENT FAILURE INFORMATION

Component Identification – 0P504C

Component Name – Emergency Service Water Pump

Pump Manufacturer – Flowserve Corporation (previously Byron Jackson Pump Company)

Pump Type – One-Stage Vertical Circulator Pump

Model No. - 24BXF 1-STG. VCT

PREVIOUS OCCURENCES

No significant previous occurrences were identified.